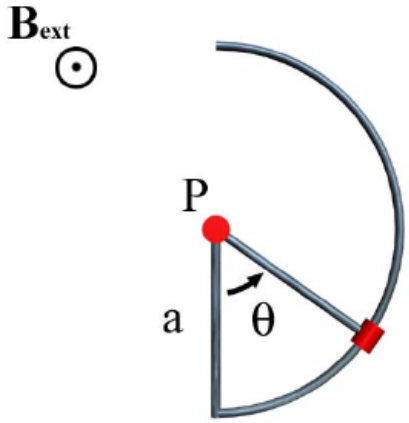
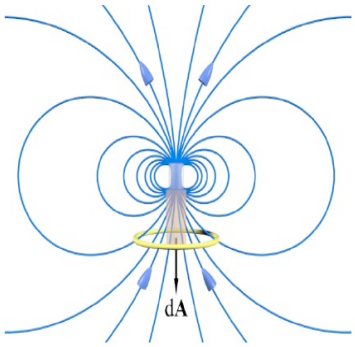
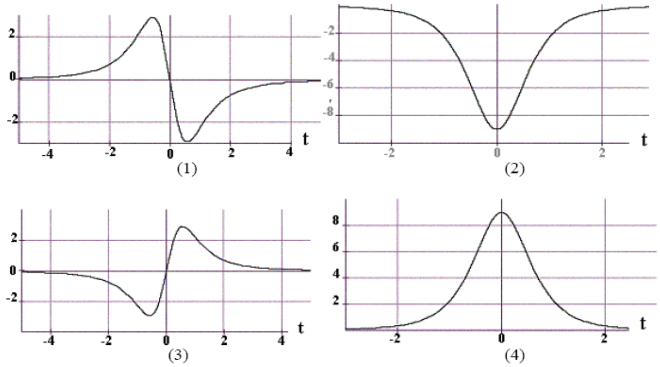
The half-circle (radius = a) conducting structure below is placed parallel to the plane of the page. The external magnetic field, $\mathbf{B}_{ext}$, is directed out of the page. The magnitude of this field varies with time as $B(t) = B_o \cos \omega t$. Point *P* is fixed, but the part between the two marks is allowed to move such that it is always in electrical contact with the rest of the structure. The angle θ (in radians) is increasing in the indicated direction such that $\theta = \omega t$ with $\theta(t = 0) = 0$.



a) Find the flux through the closed circuit at any position as a function of time.
b) Find the magnitude of the current induced in the structure at $t = t_1$. The structure has a constant resistance *R*.
c) Show the direction of the induced current on the figure.





In the figure, a permanent magnet with its North Pole pointing upward is shown. A wire loop is allowed to move from well *below* the magnet to well *above* the magnet and then back to well *below* the magnet. Which of the following is true about the curve of the magnetic flux through the wire?

a) 1 then 1 b) 2 then 2 c) 3 then 3 d) 4 then 4 e) 2 then 4 f) 4 then 2 g) 1 then 3.

The following *E* and *H* vector quantities,

$$\vec{E} = 377\, e^{-j2\,z}\ \vec{a}_x \quad , \quad \vec{H} = e^{+j2\,z}\ \vec{a}_y$$

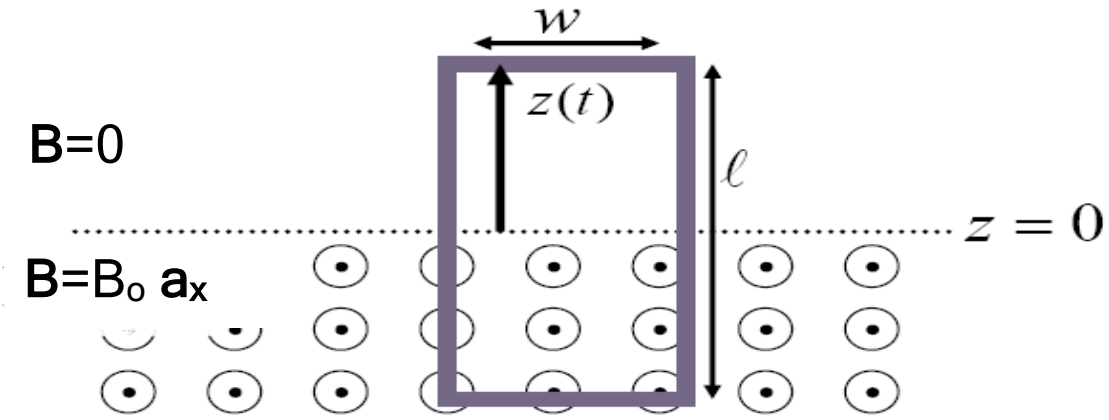
a) satisfy Maxwell's equations in free space only.
b) satisfy Maxwell's equations in all situations.
c) satisfy Newton's law of motion.
d) do not satisfy Maxwell's equations at all.
e) None of the above

Consider an infinite sheet carrying surface current density $\vec{K} = 10\ cos(10^6 t)\ \vec{a_x}$ (A/m). The sheet is parallel to the x-y plane and is placed at z = -5 cm.

Find the magnitude and direction at all times of the induced current in a conducting circular ring of radius 5 cm and resistance 10 Ω. The ring is placed in free space in the x-z plane with center at (0, 0, 10 cm).

The rectangular conducting loop below with mass m and total resistance R is placed in the region where the magnetic field is uniform below z=0 and zero above z=0, as shown. At time t=0, where exactly half of the loop is in the field region (z(t=0)=l/2), the loop is released from rest and allowed to fall due to gravity in the z-direction. Express your answers in terms of z(t).

1) Find the total flux out of the loop as a function of time t, after the loop is released.
2) What is the magnitude of the induced current in the loop after it is released?
3) What is the direction of the induced current (clockwise/counter-clockwise)?
4) What happens when the loop is completely in the magnetic field?



Consider an infinite sheet placed in the plane z = -1 m and carrying surface current density of 20 $\vec{a_x}$ mA/m.

Find the magnetic flux through the following surfaces due to this sheet.

1) (0,0,0) (0,0,1) (1,0,1) (1,0,0)
2) (1,0,1) (1,0,0) (1,1,0) (1,1,1)
